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IT18

Problem Set NO. 5

Simplex Methods

A.

1. Maximize Objective Function:

$Z = x_1 + 2x_2$

constraints

$2x_1 + x_2 \leq 8$

$x_1 + x_2 \leq 5$

$x_1, x_2 \geq 0$

Slack variables

$2x_1 + x_2 + s_1 = 8$

$x_1 + x_2 + s_2 = 5$

Objective standard form

$Z - x_1 - 2x_2 = 0$

Initial Tableau

BV	x_1	x_2	s_1	s_2	RHS
s_1	2	1	1	0	8
s_2	1	1	0	1	5
Z	-1	-2	0	0	0

2.

Maximize Objective Function:

$Z = 2x_1 + 3x_2 + 4x_3$

constraint

$x_1 + 2x_2 \leq 12$

$x_1 + x_3 \leq 8$

$x_1, x_2, x_3 \geq 0$

Slack variables

$x_1 + 2x_2 + s_1 = 12$

$x_1 + x_2 + s_2 = 8$

Objective standard form

$Z - 2x_1 - 3x_2 - 4x_3 = 0$

Initial Tableau

BV	x_1	x_2	x_3	s_1	s_2	RHS
s_1	1	2	0	1	0	12
s_2	1	0	1	0	1	8
Z	-2	3	-4	0	0	0

3. Minimize Objective Function

$W = 2x_1 + 2x_2$

Constraints

$x_1 + 2x_2 \geq 3$

$3x_1 + 2x_2 \geq 5$

$x_1, x_2 \geq 0$

convert

$x_1 + 2x_2 - s_1 + a_1 = 3$

$3x_1 + 2x_2 - s_2 + a_2 = 5$

objective

$W = 2x_1 + 2x_2 + Ma_1 + Ma_2$

standard form:

$W - 2x_1 - 2x_2 - Ma_1 - Ma_2 = 0$

BV	x_1	x_2	s_1	s_2	a_1	a_2	RHS
a_1	1	2	-1	0	1	0	3
a_2	3	2	0	-1	0	1	5
W	-2	-2	0	0	-M	-M	0

4. Minimize Objective Function

$Z = x_1 + x_2$

constraint

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$

Slack variable

$x_1 + 2x_2 + s_1 = 4$

objective

$Z - x_1 - x_2 = 0$

BV	x_1	x_2	s_1	RHS
s_1	1	2	1	4
Z	-1	-1	0	0

B.

1. Objective Function

$W = 3x_1 + 3x_2$

constraints:

$2x_1 + x_2 \geq 4$

$x_1 + 2x_2 \geq 4$

$x_1, x_2 \geq 0$

$\rightarrow$

Maximize

$Z = 4y_1 + 4y_2$

subject to

$2y_1 + y_2 \leq 3$

$y_1 + 2y_2 \leq 3$

$y_1, y_2 \geq 0$

2. Objective Function

$W = 4x_1 + x_2 + x_2$

constraints

$3x_1 + 2x_2 + x_3 \geq 23$

$x_1 + x_3 \geq 10$

$8x_1 + x_2 + 2x_3 \geq 40$

$x_1, x_2, x_3 \geq 0$

$\downarrow$

Maximize

$Z = 23y_1 + 10y_2 + 40y_3$

subject to

$3y_1 + y_2 + 8y_3 \leq 4$

$2y_1 + y_3 \leq 1$

$y_1 + y_2 + 2y_3 \leq 1$

$y_1, y_2, y_3 \geq 0$

3. Objectives Function:

$W = 9x_1 + 6x_2$

consraints

$x_1 + 2x_2 \geq 5$

$2x_1 + 2x_2 \geq 8$

$2x_1 + x_2 \geq 6$

$x_1, x_2 \geq 0$

$\rightarrow$

Maximize

$Z = 5y_1 + 8y_2 + 6y_3$

subject to

$x_1 + x_2 + 2x_3 \geq 7$

$x_1 + 2x_2 + x_3 \geq 4$

$x_1, x_2, x_3 \geq 0$

4. Objective Function

$W = 14x_1 + 20x_2 + 24x_3$

Constraints

$x_1 + x_2 + 2x_3 \geq 7$

$x_1 + 2x_2 + x_3 \geq 4$

$x_1, x_2, x_3 \geq 0$

Maximize

$z = 7y_1 + 4y_2$

$\rightarrow$ subject to:

$y_1 + y_2 \leq 14$

$y_1 + 2y_2 \leq 20$

$2y_1 + y_2 \leq 24$

$y_1, y_2 \geq 0$

C.

1. Objective Function:

$W = 10x_1 + 4x_2$

Constraints

$2x + x_2 \geq 4$

$x_1 + 2x_2 \leq 8$

$x_1, x_2 \geq 0$

variables

$2x_1 + x_2 - s_1 + a_1 = 4$

$x_1 + x_2 + s_2 = 8$

Objective

$W - 10x_1 - 4x_2 - Ma_1 = 0$

BV	x_1	x_2	s_1	s_2	a_1	RHS
a_1	2	1	-1	0	1	4
s_2	1	1	0	1	0	8
W	-10	-4	0	0	-M	0

2. Objective Function:

$W = 3x_1 + x_2 + x_4$

Constraint

$x_1 + 2x_2 + x_3 \leq 10$

$x_2 + 5x_3 \geq 6$

$4x_1 - x_2 + x_3 \geq 10$

$x_1, x_2, x_3 \geq 0$

variables

$x_1 + 2x_2 + x_3 + s_1 = 10$

$x_2 + 5x_3 + s_2 = 6$

$4x_1 - x_2 + x_3 - s_3 + a_1 = 10$

Objectives

$W - 3x_1 - x_2 - x_3 - Ma_1 = 0$

BV	x_1	x_2	x_3	s_1	s_2	s_3	a_1	RHS
s_1	1	2	1	1	0	0	0	10
s_2	0	1	5	0	1	0	0	6
a_1	4	-1	1	0	0	-1	1	10
W	-3	-1	-1	0	0	0	-M	0

3. Minimize Objective Function:

$W = x_1 + x_2$

Constraints

$2x_1 + x_2 \leq 4$

$x_1 + 3x_2 \geq 2$

$x_1, x_2 \geq 0$

variables

$2x_1 + x_2 + s_1 = 4$

$x_1 + 3x_2 - s_2 + a_1 = 2$

Objectives

$W - x_1 - x_2 - Ma_1 = 0$

BV	x_1	x_2	s_1	s_2	a_1	RHS
s_1	2	1	1	0	0	4
a_1	1	3	0	-1	1	2
W	-1	-1	0	0	-M	0

4. Minimize Objective Function

$W = 2x_1 + 3x_2$

constraints

$3x_1 + x_2 \geq 4$

$4x_1 + 2x_2 \leq 3$

$x_1, x_2 \geq 0$

variables

$3x_1 + x_2 - s_1 + a_1 = 4$

$4x_1 + 2x_2 + s_2 = 3$

Objectives

$W - 2x_1 - 3x_2 - Ma_1 = 0$

BV	x_1	x_2	s_1	s_2	a_1	RHS
a_1	3	1	-1	0	1	4
s_1	4	2	0	1	0	3
W	-2	-3	0	0	-M	0